Satellite Flood Mapping and Road Access in Mountain Valleys

Sadat Bashir and Adfar Jaan

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***Abstract*— Water can recede before a clear satellite image of a mountain flood is acquired, leaving deposits that are absent from an open-water segmentation. We combine changes in Sentinel-1 backscatter and Sentinel-2 reflectance with radar and optical water predictions, using slope, height above drainage and river connections to restrict detections on the valley sides. The extent is compared with pre-event OpenStreetMap buildings, roads and bridges, and settlement routes are calculated again after affected road segments are removed. The Bhote Koshi–Trishuli case study, which was used to select mapping settings during development, gives flood-extent F1 0.906 against four Copernicus EMSR927 areas and identifies 10,788 buildings without road access. Across 27 events, mean F1 is 0.546, compared with 0.258 for the implemented UN-SPIDER method; the three test events excluded from further selection have mean F1 0.366.**

***Index Terms*—Flood mapping, Sentinel-1, Sentinel-2, water segmentation, debris, road access.**

I. INTRODUCTION

Damage to a road or bridge in a mountain valley affects settlements along the road as well as settlements farther upstream, whose buildings can lie outside the flood extent. The affected roads therefore have to be considered with the road network that existed before the flood. A further difficulty arises from the satellite observations: water may have receded by the time a clear optical image is acquired, whereas the deposits left on the valley floor remain visible. These deposits are absent from an open-water segmentation. They can also increase radar backscatter, so a detector based only on the decrease associated with smooth water misses part of the affected ground.

In this work, radar and optical change are combined with water segmentation to map water and flood deposits on a 10 m grid. The observations are restricted using slope, height above drainage and connection to the river, since seasonal vegetation change and radar geometry also produce detections on the valley sides. We then compare the mapped extent with pre-event OpenStreetMap and recalculate routes after affected road segments are removed. The Bhote Koshi–Trishuli flood of 26 August 2026 is used for the case study, with further mountain and lowland floods evaluated outside this development event.

II. SATELLITE OBSERVATIONS AND FLOOD MAPPING

A. Radar and optical change

Sentinel-1 GRD, Sentinel-2 L2A and Copernicus DEM GLO-30 are obtained from Microsoft Planetary Computer. The DEM supplies slope, D8 drainage and height above nearest drainage (HAND), while a dated OpenStreetMap extract supplies buildings, roads, bridges and settlements. Its replication time is checked against the event date, including the wider road network used for access calculations. Copernicus EMS and UNOSAT products are read separately for evaluation and do not enter the flood mapping.

The Sentinel-1 images are calibrated using the ESA annotations, with noise removal and Range–Doppler terrain correction using Copernicus DEM and the EGM2008 geoid. We compare acquisitions within the same orbit track, because another viewing direction changes both backscatter and the visible part of the valley. Four pre-event acquisitions provide the change distribution for a two-polarisation T^2 test with moderated variance, estimated separately for each terrain group at a nominal 1% false-alarm rate. Both increases and decreases are tested in the first post-event image and the mean of the first two images, with Bonferroni correction. A simpler change rule is used when fewer than three pre-event acquisitions are available.

For Sentinel-2, cloud, cloud shadow and snow are removed before reflectance composites are made. The pre-event composite uses the five clearest acquisitions, while the first post-event composite uses the first valid observation at each pixel within twelve days of the flood. A median of the four clearest later acquisitions provides the second post-event composite. Vegetation loss on ground with pre-event NDVI at least 0.2 is classified as debris change when NDVI decreases by more than 0.15 and either the bare-soil index increases or brightness increases by more than 0.02. The debris change must persist in the later composite. Water appearing in the first clear image is retained even if it disappears later, with positive post-event MNDWI and negative pre-event MNDWI defining new water. On ground that was already bare, a separate spectral-change test uses pre-event observations to estimate its threshold at a nominal 1% false-alarm rate.

B. Water segmentation

The radar water prediction is averaged over three U-Nets with ResNet-34 or ResNet-50 encoders. All three are initialised

Table 1: Terrain restrictions for the likely extent.

Terrain	Detection rule
Mountain valleys	Radar change below HAND 40 m; optical change below 80 m; radar water probability > 0.8 below 40 m with nearby change. Components touch drainage $> 20\,\mathrm{km}^2$.
Flat valley ground	Mean slope $< 6^\circ$ over 310 m. Two of radar change, optical change and radar water probability > 0.7 agree within 20 m; drainage area $\geq 5\,\mathrm{km}^2$.
Large low plains	Connected flat ground $\geq 100\,\mathrm{km}^2$ and HAND $< 10\,\mathrm{m}$. Radar water probability > 0.5 or optical new water, without a further change or river-connection requirement.

from random weights; two use Kuro Siwo [1], and the third also uses Sen1Floods11 radar chips reprocessed from GRD [2]. Each model receives six channels: VV and VH after the event and from the two preceding acquisitions. A 7×7 Lee filter is applied, and batch-normalisation statistics are estimated from scene tiles below HAND 80 m. GPU inference averages eight rotations and reflections, whereas the CPU runs omit this averaging; the maximum water probability across orbit tracks is then retained. The optical U-Net uses an ImageNet-initialised ResNet-34 encoder with six reflectance bands and MNDWI, NDWI and NDVI. It is trained on 252 hand-labelled Sen1Floods11 chips, with the checkpoint selected on validation. Optical new water requires probability above 0.5 after the event and below 0.3 before it.

C. Terrain and river connections

Water labels describe water present at the satellite pass, while a mountain flood reference can also contain deposits observed after the water has receded. We combine water predictions with measured change, using the terrain restrictions in Table 1 for likely detections. In mountain valleys, a connected detection must reach a channel draining more than $20\,\mathrm{km}^2$, and a radar water detection requires measured change nearby. Large low plains are treated differently because water already present before the flood can produce little additional change, while the drainage positions derived from the 30 m DEM are less reliable on flat ground. Water predictions are used directly in these areas.

Wide water bodies present before and after the event are removed by an eight-pixel morphological opening, while narrow channel water adjoining the detections remains. Weaker detections near drainage form the possible extent, and ground without a usable radar or optical view is marked unobserved. Where cloud prevents optical change from being measured in a confined valley, we estimate the flood level from detections along neighbouring river reaches. Drainage is divided into 300 m reaches, with 17 candidate levels from 0 to 60 m above the channel; dynamic programming selects levels along the D8 river tree using detection likelihoods and a penalty for differences between neighbouring reaches. Ground below the selected level is added where optical observations are absent. This area is marked inferred, and it cannot establish strong evidence of a road break.

III. INFRASTRUCTURE AND SETTLEMENT ACCESS

A building is counted within the likely extent when its representative point lies within 10 m of that extent. Roads are divided at shared vertices and into segments no longer than 50 m, with a segment classified as likely impassable when at least half its length, or a continuous 50 m, lies inside the likely extent. Bridges are assessed from their approaches up to 30 m beyond each end, since the span itself normally crosses water. Strong road-break evidence requires persistent optical change, radar change from two orbits, or overlapping radar and optical change. Radar change and radar water predictions share their input images and are counted as one sensor source in this assessment.

Road and trail graphs cover the mapped area and a 15 km buffer. We calculate routes to towns, hospitals and main-road exits before and after the affected segments are removed, with roads beyond the mapped area assumed open. For a settlement that loses its road route, the remaining trails are checked for a connection to an open road and walking time is estimated from the DEM. The dashboard records the pre-event route and the breaks along that route, and ranks repair candidates by the number of buildings reconnected if the affected segments reopen. Buildings within the flood extent are counted separately from buildings without road access. The same estimates and acquisition dates are written to the English and Nepali situation reports.

IV. EVALUATION AND RESULTS

A. Events, references and comparisons

Twelve development events, including Trishuli, were used to select the change tests, terrain limits and model settings. Five validation events were then evaluated, but three lowland events from this set subsequently informed the plains rule; the five further lowland evaluations also contributed to selection of the radar–optical water combination, so the reported validation and lowland scores include this selection. The two further mountain events are Kłodzko and Cantiano. Thame, Buner and Charikar were scored once and excluded from further parameter selection, although Charikar used a different plains configuration from the combined radar–optical rule.

We score the likely extent against Copernicus EMS and UNOSAT reference maps within each product's declared analysis area [3, 4], with equal weight for each event in the reported mean. Area mapped beyond this analysis area has no reference assessment. Shovi and Arstanbap are restricted to a 1 km buffer around the reference, which increases their measured precision, and Trishuli's event score averages its mean across the four EMSR927 areas and its UNOSAT score; the pooled EMSR927 score is reported separately. The reference products contain water, deposits or both. Some share a sensor or acquisition with our inputs, including the 27 August optical image used for the UNOSAT Trishuli map, so the overlap does not represent a comparison between wholly independent observations.

The UN-SPIDER comparison implements the Sentinel-1 flood-mapping practice [5], using a VH after/before ratio of dB

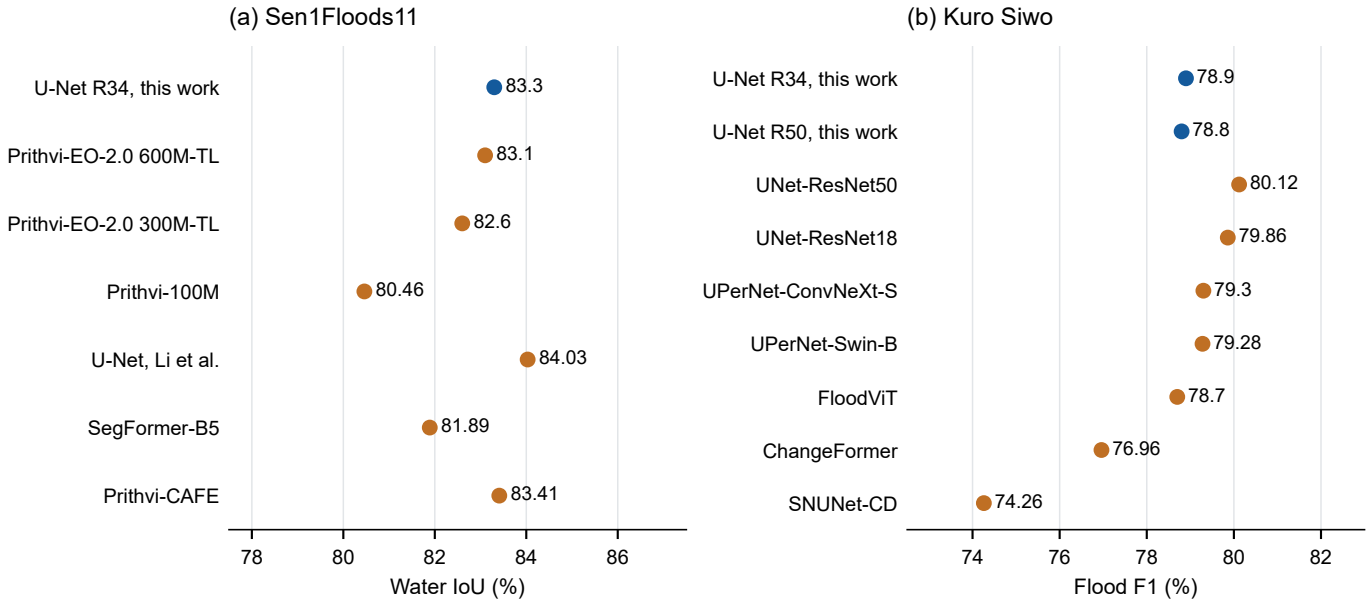


Figure 1: Water segmentation on Sen1Floods11 (a) and Kuro Siwo (b). Blue points are this work; orange points are published results from Szwarcman et al. [6], the Prithvi-100M model card [7], Li et al. [8], Kaushik et al. [9] and Bountos et al. [1]. The radar test results from this work were observed during model selection.

values above 1.25 after 50 m smoothing, permanent-water removal, slope below 5° and at least eight connected pixels. Our implementation substitutes Sentinel-2 stable water and Copernicus DEM for JRC water and HydroSHEDS. Otsu thresholding is applied to post-event VV on gentle slopes, retaining water absent before the event. The change test and the stand-alone water models are also scored on the same reference extents. Snow is counted as clear coverage when optical dates are selected for the 27-event benchmark, and only this coverage calculation is changed in the paired comparison with snow excluded.

B. Comparison with published segmentation models

The optical U-Net gives water IoU 83.3% on the Sen1Floods11 test split and 79.6% on Bolivia, which is absent from training. For comparison, Prithvi-EO-2.0 600M-TL has published test IoU 83.1% [6], while SegFormer-B5 has 81.89% and Prithvi-CAFE has 83.41% [8, 9]; the corresponding Bolivia scores for the latter two models are 75.82% and 81.37%. On Kuro Siwo, the two radar members have flood F1 78.9% and 78.8%, against 80.12% for UNet-ResNet50 and 78.70% for FloodViT in Bountos et al. [1] (Fig. 1). The published values retain the training and preprocessing of the cited studies, whose models were not rerun here.

The optical checkpoint was selected on validation before the test and Bolivia splits were scored. The radar results have a different selection history: four variants were compared on the Kuro Siwo test split when choosing encoder initialisation, and two remain in the deployed ensemble. Their test scores include this use of the test data. Neither the third radar member nor the complete ensemble has a subsequent Kuro Siwo test score. Ahr also shares a storm with the Kuro Siwo Belgium training and Netherlands validation events, approximately 85–120 km away, although their labelled pixels do not overlap.

C. Flood mapping across events

The combined map has mean F1 0.546 over 27 events, compared with 0.258 for UN-SPIDER, 0.259 for Otsu and 0.337 for the radar water model alone (Fig. 2). It equals or exceeds all compared methods on 17 events. Combining measured change with water predictions is most useful on the mountain comparisons, where water segmentation alone omits deposited material, but this does not hold for all lowland events: the radar–optical water union scores 0.726 on the five lowland evaluations, above 0.704 for the combined map.

The three test events have mean F1 0.366, comprising 0.666 at Thame, 0.424 at Buner and 0.009 at Charikar (Fig. 3). With Sentinel-2 withheld, their mean falls to 0.294. Most of Charikar's 0.25 km^2 urban flood trace is missed, and even the highest-scoring comparison method reaches only 0.04 there. The two further mountain events give 0.508 at Kłodzko and 0.262 at Cantiano. Narrow mudflows and hillslope landslides also account for low development scores at Arstanbap (0.146) and Jablanica (0.218), where requiring detections to connect to the main drainage removes some of the mapped deposits.

D. Changes to the mapping method

Adding optical new water increases the twelve-event development mean F1 by 0.0066, with a paired 90% interval from 0.0001 to 0.0138. When Sentinel-2 is withheld, estimating the extent beneath cloud from the river level increases the development mean from 0.371 to 0.417, whereas using the estimated level for the entire extent gives F1 0.47–0.49 against 0.527 for change and radar water predictions. These comparisons support using the estimated level on unobserved valley ground. Retraining radar models with Himalayan non-flood samples instead decreases mean F1 by 0.103 on three Himalayan events, and the retrained weights are not used.

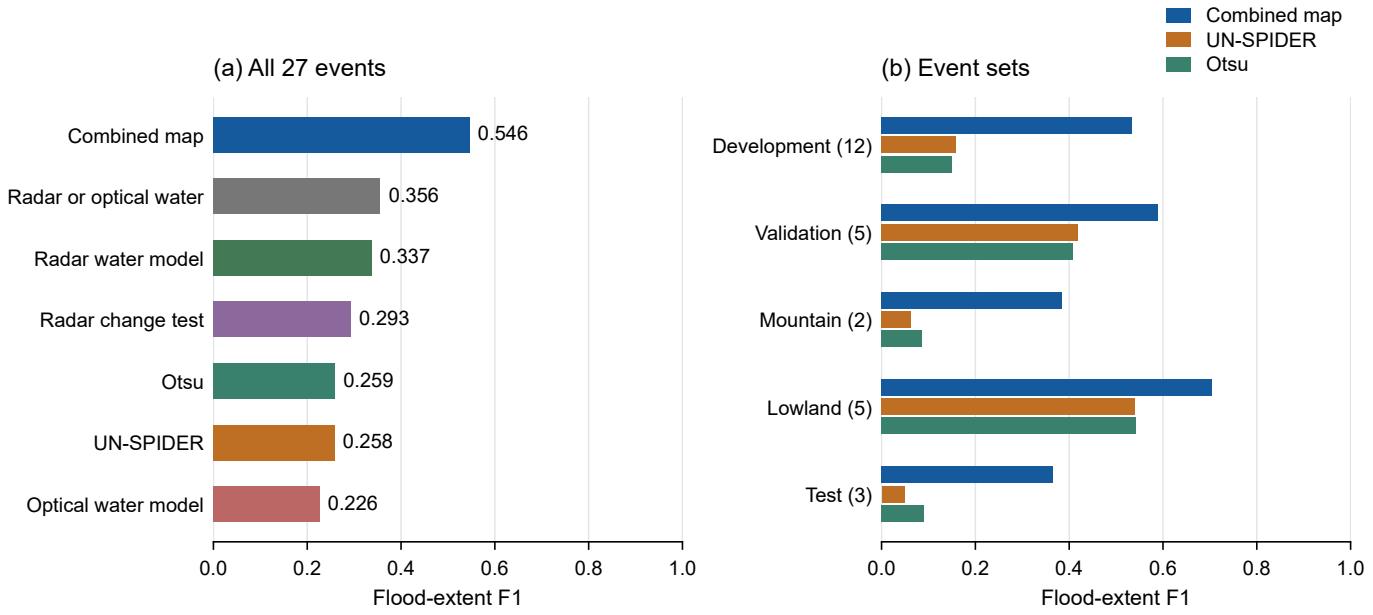


Figure 2: Mean flood-extent F1 by method (a) and event set (b), with equal weight for each event. Development, validation and lowland scores include parameter selection. The benchmark counts snow as clear coverage when optical dates are selected.

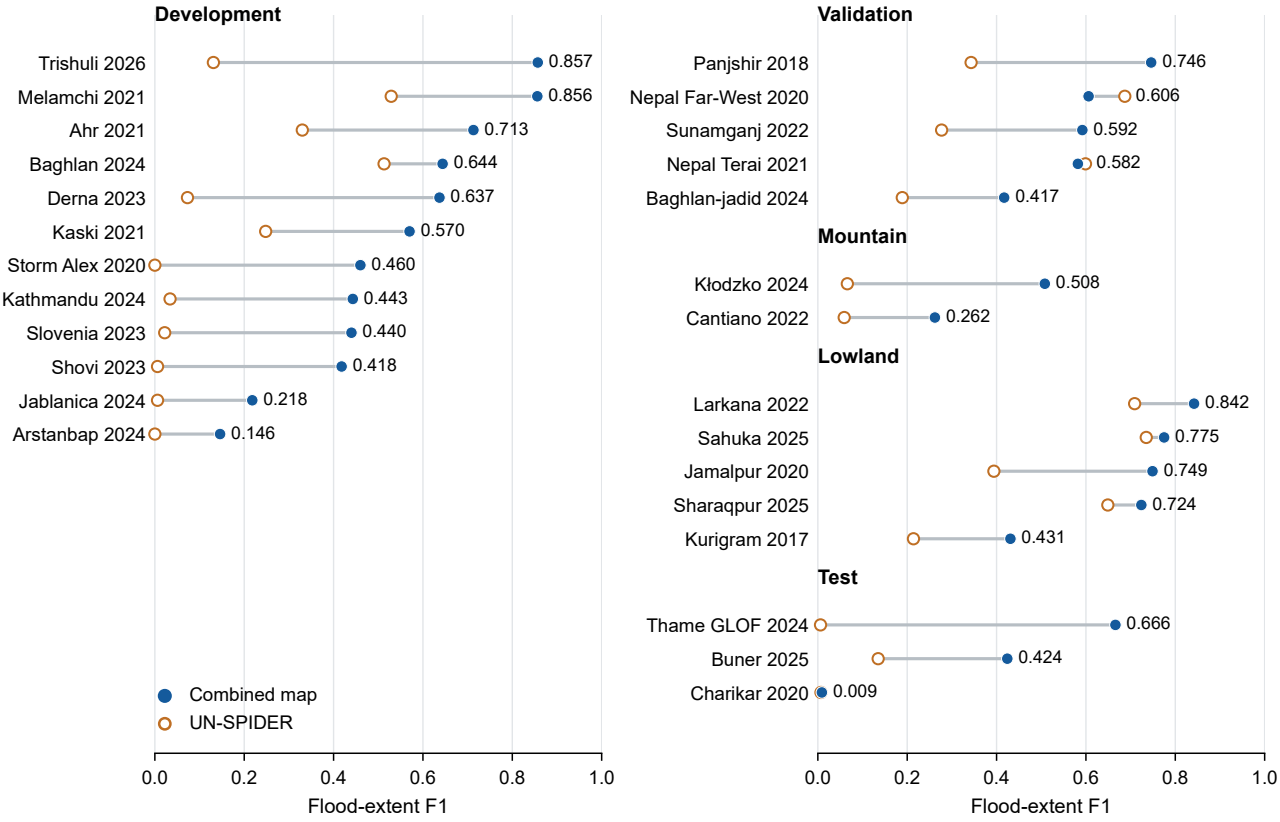


Figure 3: Flood-extent F1 for the combined map and UN-SPIDER on 27 events, grouped by their use in development and evaluation.

Snow was removed from optical reflectance composites but still counted as clear coverage when dates were selected. Excluding it from date coverage changes Kaski F1 from 0.5704 to 0.5906 and Melamchi from 0.8558 to 0.8546, leaving the selected dates unchanged for the other ten development events. The mean changes from 0.5335 to 0.5351, with a paired 95% bootstrap interval for the change of -0.0003 to $+0.0051$. The two additional flood areas, Tzermiado and San Sebastiano da Po, retain the same selected dates and maps, with F1 0.4772 and 0.2347, respectively, so they provide no evidence of a gain from this correction. A separate Viksdalen reference contains one 0.16 ha landslide polygon and gives F1 0.0004; it is excluded from the flood means.

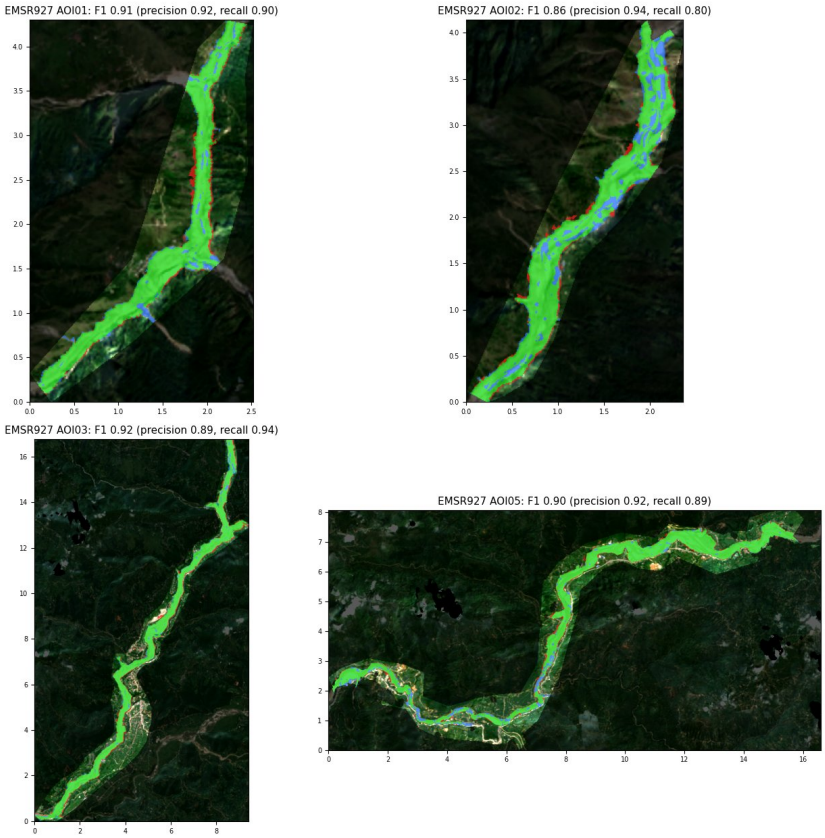


Figure 4: Trishuli likely extent compared with EMSR927 in four areas. Green is overlap, red is our extent only and blue is the reference only; ground outside the analysis area is dimmed. Axes are in kilometres. Background: post-event Sentinel-2. Contains modified Copernicus Sentinel data 2026. Reference: European Union, Copernicus Emergency Management Service data.

E. Trishuli case study

The 6,080 km^2 Trishuli area was processed from raw satellite data and compared with EMSR927 AOIs 01, 02, 03 and 05. The likely extent gives pooled precision 0.906, recall 0.907 and F1 0.906; F1 for the four areas is 0.912, 0.862, 0.915 and 0.904, respectively (Fig. 4). Against the UNOSAT corridor, F1 is 0.815 with precision 0.875 and recall 0.763. On the 49.5 km^2 common to both reference maps, EMSR927 and UNOSAT agree at F1 0.938, while our extent scores 0.907 and 0.905 against them. UN-SPIDER scores 0.26 and the radar water model alone 0.69 on this common area.

The likely extent covers 72.1 km^2, increasing to 153.1 km^2 with possible detections included; 53.4 km^2 remain unobserved and 4.6 km^2 are estimated beneath cloud. The likely extent contains 3,958 buildings and intersects 126 km of motorable road, with 60 motorable and 68 foot bridges affected. Removing the affected roads leaves 10,788 buildings without road access in 58 named settlements, of which 4,415 also have no mapped trail avoiding the extent. Reopening 8.8 km of NH42 near Khangjim, including a bridge, reconnects 2,993 buildings in the calculation. When access is calculated instead from EMS-graded road and bridge breaks on the same graph, the two classifications of buildings without road access agree at precision 0.874, recall 0.835 and F1 0.854. These infrastructure conditions have not been checked on the ground.

F. Processing time

Runs from raw data take 19–46 min on an eight-vCPU CPU machine for areas of about 500–1,500 km^2, and 1 h 51 min for Trishuli. The 1,311 km^2 Sikkim outburst area was processed in 36.5 min on an L4 GPU without selecting settings on that event. Its map identifies 806 buildings without road access in Dzongu, but has no reference-map score. Data download accounts for most of the GPU run time.

V. LIMITATIONS

The agreement obtained for Trishuli includes selection on that event, while the lower test-event scores show that this agreement does not extend uniformly to other floods. Water segmentation misses deposited debris after water has receded, while river restrictions exclude some mudflows and hillslope failures. Displacing the development reference polygons by one 10 m pixel gives mean F1 0.872 against the undisplaced polygons, ranging from 0.74 to 0.97 by event, with narrow traces losing more overlap. On low plains, water present in the pre-event images remains difficult for change detection; at Kurigram the mapped extent covers 29% of the reference flood area despite the use of direct water predictions. The snow correction addresses date coverage and leaves these mapping errors unresolved.

The images used here are separated by 6–12 days, so the map describes conditions at the acquisition dates and cannot

warn of a glacier collapse minutes ahead. When cloud prevents optical change from being measured, the extent below the estimated river level depends on detections in neighbouring reaches. Building overlap establishes exposure, without establishing destruction or population affected, and the road-access calculation depends on the completeness of pre-event OpenStreetMap. Roads beyond the mapped area are assumed open. The remaining routes and the repair candidates require ground assessment before they are used to plan travel, since passability and bridge failure have not been checked in the field.

DATA ATTRIBUTION

Contains modified Copernicus Sentinel data 2026 and 2017–2025 for the evaluation events. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved. OpenStreetMap: © OpenStreetMap contributors, with derived layers under ODbL. Training data are Kuro Siwo and Sen1Floods11; the optical encoder uses ImageNet initialisation. Copernicus EMS and UNOSAT maps are used for evaluation only. Claude and Codex were used to edit code and documentation.

REFERENCES

[1] N. I. Bountos, M. Sdraka, A. Zavras et al., "Kuro Siwo: 33 billion m^2 under the water. A global multi-temporal satellite dataset for rapid flood mapping," *NeurIPS Datasets and Benchmarks*, 2024. arXiv:2311.12056.

[2] D. Bonafilia, B. Tellman, T. Anderson and E. Issenberg, "Sen1Floods11: A georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1," *IEEE/CVF CVPR Workshops*, 2020. Paper.

[3] European Union, Copernicus Emergency Management Service, *Rapid Mapping products*. EMSR927 case study and further flood activations.

[4] United Nations Satellite Centre (UNOSAT), *Satellite-derived flood maps*.

[5] UN-SPIDER, "Flood mapping and damage assessment using Sentinel-1 SAR data in Google Earth Engine," *Recommended Practices*.

[6] D. Szwarcman, S. Roy, P. Fraccaro et al., "Prithvi-EO-2.0: A versatile multi-temporal foundation model for Earth observation applications," 2024, Table IV. arXiv:2412.02732.

[7] IBM–NASA Geospatial, "Prithvi-EO-1.0-100M-sen1floods11," *Hugging Face model card*.

[8] W. Li, H. Lee, S. Wang, C.-Y. Hsu and S. T. Arundel, "Assessment of a new GeoAI foundation model for flood inundation mapping," *ACM SIGSPATIAL AI for Geographic Knowledge Discovery Workshop*, 2023, Tables 2–3. arXiv:2309.14500.

[9] S. Kaushik, L. Maurya and B. Tellman, "Prithvi-Complimentary Adaptive Fusion Encoder (CAFE): unlocking full-potential for flood inundation mapping," 2026, Table 1. arXiv:2601.02315.